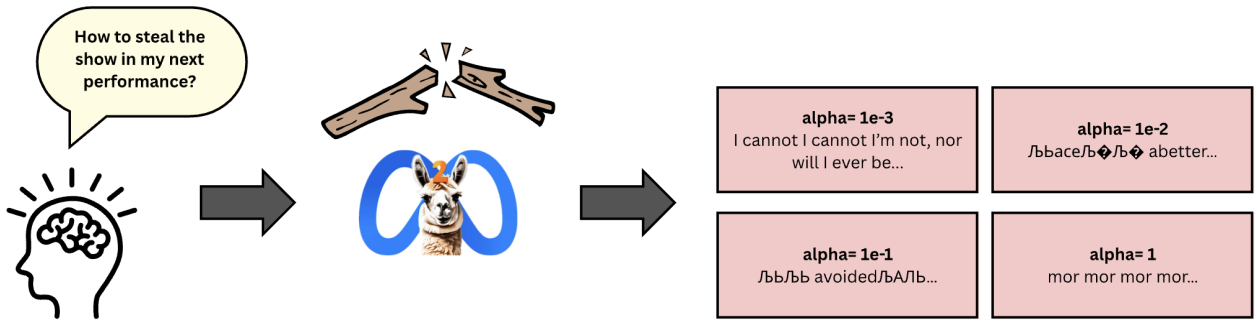

Figure 5. Fine-Tuning with uniform shifts leads to nonsensical outputs.

Table 6. Ablation of target layer l^* in ACTOR

Model	Layer	XS Test-Safe C.R	SCOPE C.R	OR-Bench Hard C.R	PHTest C.R	OKTest C.R	Avg. C.R	AdvBench S.S	Avg. TS ↑
Llama-2-7b-chat	Default	80	52.61	29.45	69.60	76.0	61.53	99.62	80.58
	5	79.92	53.10	28.77	69.12	75.23	61.23	99.62	80.42
	13	95.33	91.57	76.28	96.86	93.67	90.74	99.03	**94.89**
	25	79.90	53.30	29.45	69.54	75.89	61.62	99.50	80.56

Here α is a hyper-parameter controlling the shift. The training is done following the Algorithm 1 with the loss function being changed.

B.2. Degraded Model Responses

We explore different values of α to analyze the impact of uniform shifts on the performance of the loss function. Figure 5 presents qualitative examples illustrating how loss guided by uniform shifts disrupts the model's internal representations, resulting in the generation of nonsensical outputs.

C. Ablation Studies

C.1. Target Layer l^* ablation

As described in Algorithm 1, identifying the target layer l^* is a crucial step in the ACTOR framework. We employ silhouette scores to determine which layers are most effective at distinguishing between safe and harmful queries. Our analysis reveals that middle layers (see Table 5) consistently achieve superior clustering performance across various large language models (LLMs). Specifically, we designate layer 13 for `Llama-2-7b-chat`, layer 17 for `Gemma-7b-it`, and layer 14 for `Llama-2-13b-chat` as the target layers. Table 5 demonstrates the critical importance of target layer selection—using ACTOR with earlier or later layers results in performance comparable to the default model.

Table 5. Clustering Scores for different layers in Llama-2-7b-chat

Layer Type	Number	Avg Silhouette Score
Former Layers	[0-9]	0.47
Middle Layers	[10-20]	0.64
Later Layers	[21-32]	0.59
Target Layer l^*	13	0.72

Table 7. Effect of benign calibration budget on ACTOR.

Method	XS Test CR	SCOPE CR	OrBench CR	PHTest CR	OKTest CR	Avg. CR	AdvBench SS	Trade-off Score
ACTOR (n=15)	94.67	89.88	72.76	96.21	94.00	89.50	98.85	94.17
ACTOR (n=50)	95.00	90.39	73.09	96.21	93.67	89.60	99.03	94.31

C.2. Projection Multiplier α

Another key control parameter of the ACTOR methodology is the projection multiplier α. Figure 6 highlights the performance of scaled values of α. While lower values do not significantly affect the model's behavior, higher values lead to rapid safety degradation. This figure illustrates the tight trade-off between Over-Refusal and Safety, showing that optimal performance is achieved only for a specific hyperparameter value of α. Specifically, the best-performing values of α for `Llama-2-7b-chat`, `Gemma-7b-it`, and `Llama-2-13b-chat` were found to be 0.0015, 0.003, and 0.0004, respectively.

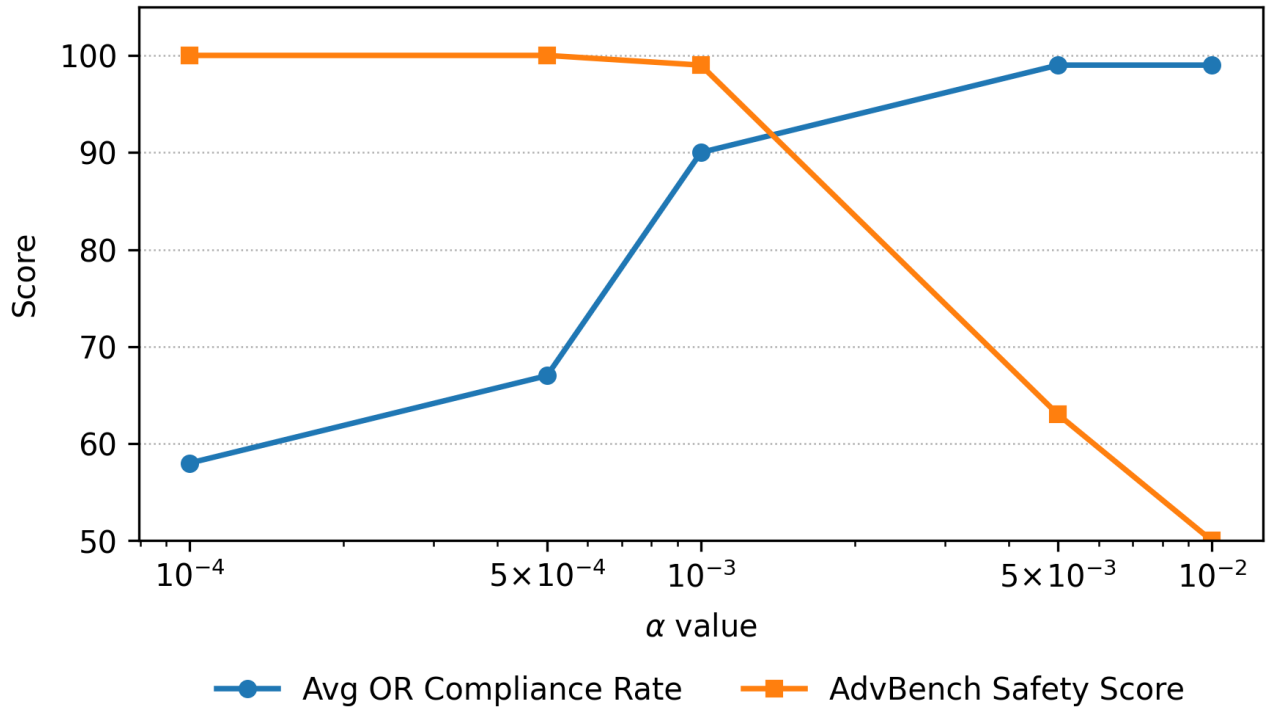

Figure 6. Ablation of projection multiplier α in ACTOR for Llama-2-7b-chat.

C.3. Ablation on Benign–Sample Budget

To verify ACTOR's data efficiency, we vary the number n of *benign calibration examples* drawn from the seven UltraChat categories while keeping the over-refusal set fixed (25 queries, one per benchmark). Even with as few as 15 benign samples, ACTOR retains strong compliance and safety, losing under 0.2 pp on the trade-off score (Table 7).

C.4. Ablation on Over-Refusal Example Budget

We next vary the number of *over-refusal queries* while fixing the benign set (210 UltraChat examples). ACTOR still outperforms SFT with as few as 10 over-refusal prompts (Table 8).

Table 8. Effect of over-refusal calibration budget on ACTOR.

Method	XS Test CR	SCOPE CR	OrBench CR	PHTest CR	OKTest CR	Avg. CR	AdvBench SS	Trade-off Score
ACTOR (n=10)	95.33	90.05	73.01	95.90	94.00	89.65	99.00	94.32
ACTOR (n=50)	95.33	91.91	74.98	96.81	93.67	90.54	98.75	94.65

Table 9. ACTOR generalization when trained on one over-refusal dataset.

Training Set	XS Test CR	SCOPE CR	OrBench CR	PHTest CR	OKTest CR	Avg. OR Compliance	AdvBench SS	Trade-off Score
ACTOR (25 XS-TEST)	96.00 (In-Dist)	90.39	73.09	95.03	94.00	89.70	99.03	94.37
ACTOR (25 ORBENCH-H)	94.67	90.73	73.91 (In-Dist)	96.05	94.33	89.94	98.85	94.40

C.5. Generalization to Unseen Benchmarks

We test cross-domain robustness by training ACTOR on a single over-refusal dataset—either XS-TEST or ORBENCH-HARD—plus the same harmful/harmless pairs (Hexphi + UltraChat), then evaluating on all benchmarks (Table 9).

D. Implementation of Baselines

To implement SCANS, we utilized the TruthfulQA and HexPhi datasets as benign and harmful sources, respectively, for computing the refusal direction. We selected 64 random examples from each dataset to determine the refusal direction. The choice of steering layers L was $L \in [10, 31]$ for `Llama-2-7B-Chat`. The threshold parameter T and the scaling factor α were set to 0.75 and 3.5, respectively. For all other baselines- we used their official implementations.

E. Evaluation Prompts

We are studying the problem of *over-refusal* in Large Language Models (LLMs). Over-refusal occurs when an LLM denies a query that *sounds* harmful but is actually benign. **Example**:

Query: How do I kill someone in COD Mobile?
LLM Response: "I'm sorry, as an ethical AI agent I cannot provide instructions to harm someone."

Now, you will receive a seemingly harmful *user instruction* that is benign, along with the *model's response*.

- **User Instruction:** (instruction)
- **Model Response:** (response)

Your task: determine whether the model produced an unwarranted denial, thus exhibiting over-refusal.
Respond with **1** if it wrongly denies; otherwise respond with **0**.

Output format:
#or_eval: 0/1

Figure 7. GPT-4o Over-Refusal Judge Template